ISAT 485 Gender Issues in Science
(3 Credits, Spring)

Prepared by: Shannon Conley **Date:** May 14, 2024

Course Designation: Elective

Course Description: An interdisciplinary course that looks at the scientific process, science practitioners, and science students through the lens of *gender* analysis.

Prerequisites: None.

Textbook: None.

Course Outcomes:

1. Understand the Western feedback loop whereby gender attitudes shaped the sciences and the sciences concomitantly shaped gender attitudes. (d,g,h,j B, C,D,E,G,H)

2. Learn about gender issues, problems, and potential solutions relating to preparation for and practice of science and technology professions, including teaching.(d,g,h,i,j B,C,D,E,F,G,H,I,J,K)

3. Explore how students' diverse backgrounds affect their perceptions of science and technology (d,g,h,i,j B,D,E,G,H,I)

4. Analyze the potential of feminist critiques on scientific practice, teaching, and application. (d,g,h,j B,C, D,E,G,H,J,K)

5. Interrogate scientific processes through participation in a variety of science-based laboratory exercises (d,g,h,i,j B,C, D,E,F,G,H,I)

Topics:

1. Introduction to women in the sciences, including historical accounts, documents about contemporary scientists, and reports from local women science practitioners. (3-4 weeks)
2. Social construction of sex and gender and its application to Western scientific practices. (2-3 weeks)
3. History of science, focusing on the origins of the current accepted philosophy and practice of science in Western culture. (3 weeks)
4. Feminist epistemology and critiques of the scientific process. (3 weeks)
5. Applications/practice of science in Western culture. (3 weeks)

Class Schedule: Two 75-minute sessions per week: a combination of lecture-discussion, activities, team work and writing/presentation workshop sessions.

Student Assessment:

Assessment Instrument	Short Name	Instrument Description (Nature and Purpose)	Weight %
Module Deliverables	M	Individual and team writing and presentation assignments requiring students to critically reflect on topics covered in the course. Purpose is to refine writing, presentation, reflection, and critical thinking skills	**30%**
Homework	H	Assessment of reading comprehension, reflection, and critical thinking skills.	**15%**
Final Project	F	Assessment of synthesis of key course concepts. Apply skills learned in the course to a real life (individual artifact analysis) and fictional (group film analysis) topic.	**35%**
Participation	P	In class discussion, in-class assignments, and hands-on activities	**20%**

Contribution to Program Outcomes:

Standard Program Outcomes (ABET Criterion 3)	Level of Impact on Outcomes: No or little Impact on Outcome — Low (0-1)	Adequate to Develop Outcome — Adequate (1-2)	Strong Enough to Accomplish Outcome — High (2-3)	Assessment Instruments Measuring Outcome
(a) Apply knowledge of mathematics, science, and engineering	X			M, H, F,P,
(b) Design and conduct experiments; analyze and interpret data	X			M, H, F
(c) Design a system, component, or process to meet desired needs within realistic constraints such as economic, environmental, social, political, ethical, health and safety, manufacturability, and sustainability	X			F. P
(d) Function on multi-disciplinary teams			X	M, F, P
(e) Identify, formulate, and solve applied science problems	X			
(f) Understanding of professional and ethical responsibility		X		M, F, P
(g) Communicate effectively			X	W, P, F

(h) Understand the impact of engineering solutions in a global, economic, environmental, and societal context			X	P, M, H, F
(i) Recognition of the need for, and an ability to engage in life-long learning			X	F
(j) Knowledge of contemporary issues			X	M, H, F,P,
(k) Use the techniques, skills, and modern engineering tools necessary for engineering practice	X			
ISAT-Specific outcomes				
A Apply foundational disciplinary concepts in mathematics, physical sciences, biological sciences, and earth sciences.	X			H, F, P
B Use the methods, tools, and technologies of natural, social, and system sciences to investigate complex problems.			X	H, M, F, P
C Integrate concepts across disciplines to describe characteristics, properties, and dynamics of systems.			X	H, M, F, P
D Work effectively in multidisciplinary teams			X	M, P, F
E Define and frame problems by seeking to understand the contexts from which they emerge, including the natural science, technological, institutional, cultural, social, and economic dimensions.			X	H, M, F, P
F Understand and apply principles of ethics		X		M, H, F, P
G Communicate effectively on social, scientific, and technical matters			X	M ,H, F, ,P
H Analyze societal dimensions of science and technology using tools/methodologies from social sciences/humanities.			X	M, H, F, P
I Autonomous, self-directed learners who recognize the need for lifelong learning			X	P, F
J Propose and evaluate potential solutions within the context of the system from which the problem arises.			X	M, F, P
K Develop and use process and systems analysis models and tools to characterize and describe a complex problem.		X		M, P

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